Algoritmos

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R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see http://rmarkdown.rstudio.com.

When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

Primeros comandos en R

R es un lenguaje muy similar a matlab y permite el trabajo con matrices.

```
 A=38
 A=40
 B=60
 C=B-A
 C
```

```
## [1] 20
```

```
Y=40
alpha<-30
```

Uso de datasets (base de datos internas de r)

Usando el comando data en la consola obtenemos distintos datos ya crgados en la base de datos de R, estos datos están dados en las tables las cuales podemos elegir columnas/filas específicas (escribiendo el el signo $)

```
summary(cars)
```

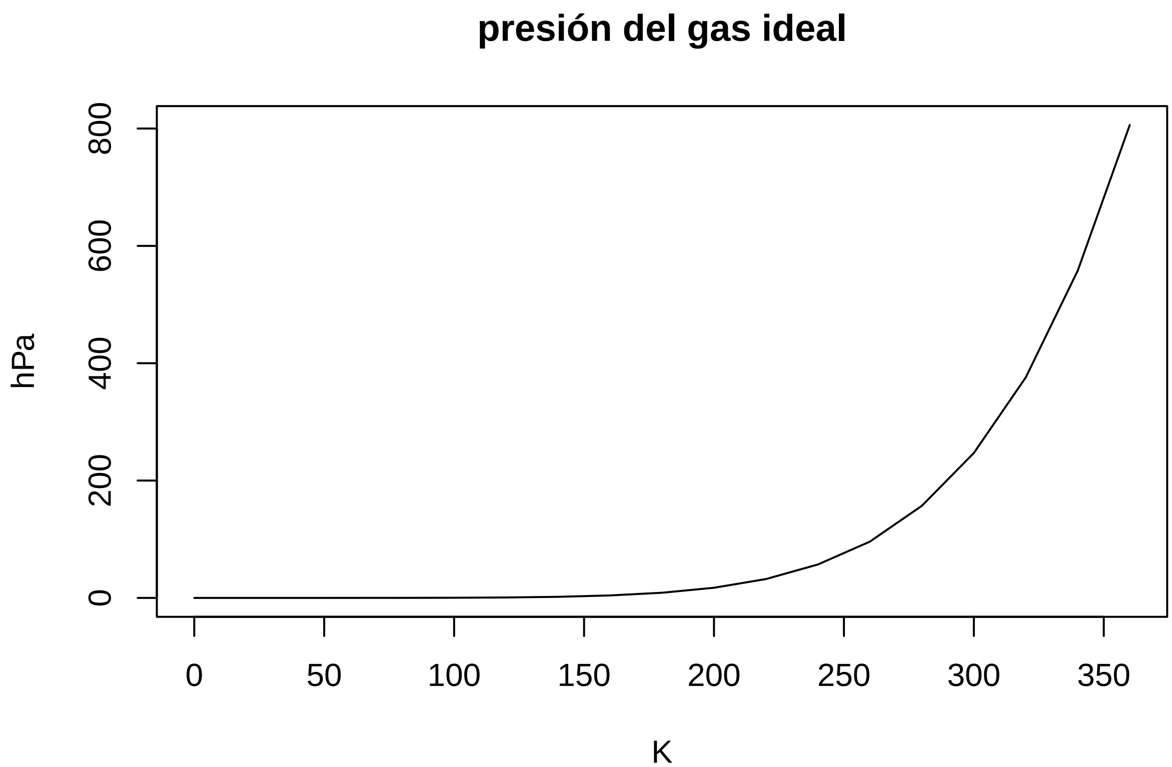
```
##      speed           dist       
##  Min.   : 4.0   Min.   :  2.00  
##  1st Qu.:12.0   1st Qu.: 26.00  
##  Median :15.0   Median : 36.00  
##  Mean   :15.4   Mean   : 42.98  
##  3rd Qu.:19.0   3rd Qu.: 56.00  
##  Max.   :25.0   Max.   :120.00  
```

Including Plots

Si no sabemos para que sirve una función usamos "?" para pedir información sobre la función (ej: ?plot)

You can also embed plots, for example:

```
plot(pressure,type="l",main="presión del gas ideal",ylab="hPa",xlab="K")
```



Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot. Le sacamos el `echo = FALSE`. Plot es un comando que me permite graficar cualquier fuente de datos que le asignemos, teniendo también la posibilidad de asignar nombre a los ejes o hacer cambios al gráfico.

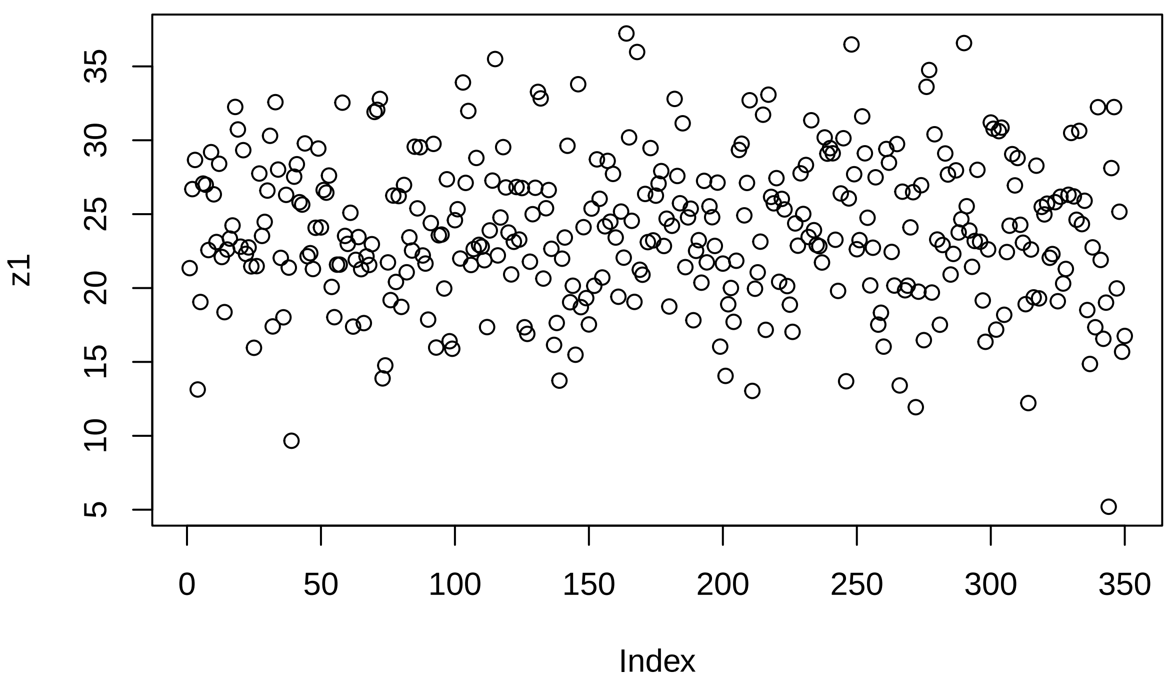
Funciones estadísticas

```
z1<-rnorm(350,24,5) #renorm genera 350 número aleatorios con promedio 25 y desvío estándar de 5
z1
```

```
##   [1] 21.346962 26.692993 28.666938 13.132822 19.054925 27.068143 26.987896
##   [8] 22.569418 29.194950 26.339579 23.112628 28.407298 22.097653 18.368575
##  [15] 22.605624 23.344048 24.233827 32.254407 30.729510 22.798664 29.328333
##  [22] 22.312142 22.748146 21.465614 15.955629 21.481865 27.747957 23.528073
##  [29] 24.476357 26.593776 30.302586 17.400805 32.578018 28.016098 22.040896
##  [36] 18.018411 26.305525 21.380886  9.663066 27.534950 28.376639 25.807083
##  [43] 25.647092 29.786155 22.147658 22.358604 21.287185 24.077723 29.440361
##  [50] 24.100228 26.631053 26.453787 27.611184 20.074682 18.027193 21.583303
##  [57] 21.584774 32.545348 23.527054 22.988115 25.092206 17.390684 21.907210
##  [64] 23.448051 21.275847 17.625772 22.137218 21.561713 22.965702 31.917084
##  [71] 32.059274 32.798492 13.885246 14.763083 21.732692 19.183167 26.256503
##  [78] 20.411113 26.223243 18.723008 26.975410 21.065619 23.429828 22.536208
##  [85] 29.562808 25.390769 29.526544 22.203246 21.655100 17.862676 24.397268
##  [92] 29.754504 15.971773 23.569010 23.616023 19.967661 27.358162 16.400871
##  [99] 15.891589 24.589765 25.331342 21.994201 33.909590 27.115062 31.981820
## [106] 21.560136 22.639756 28.803203 22.922997 22.794278 21.881865 17.358042
## [113] 23.896124 27.270773 35.495925 22.205492 24.774963 29.534463 26.805332
## [120] 23.753742 20.920703 23.129156 26.835285 23.281178 26.766233 17.344639
## [127] 16.896032 21.781422 24.996441 26.783439 33.273128 32.830626 20.639490
## [134] 25.399171 26.630711 22.653836 16.152443 17.636247 13.735097 21.985364
```

```
## [141] 23.417248 29.626895 19.038398 20.154682 15.486447 33.791936 18.706049
## [148] 24.116297 19.318003 17.536867 25.377595 20.150890 28.705872 26.038538
## [155] 20.712442 24.177103 28.601976 24.489664 27.722870 23.416745 19.411371
## [162] 25.172895 22.056741 37.225515 30.194667 24.548878 19.062049 35.972375
## [169] 21.235590 20.900318 26.367087 23.106724 29.471675 23.222110 26.251401
## [176] 27.066392 27.913556 22.845735 24.691612 18.750419 24.213034 32.797771
## [183] 27.581876 25.736002 31.147159 21.413938 24.779283 25.370294 17.819266
## [190] 22.517131 23.239983 20.362170 27.251860 21.735256 25.530184 24.790003
## [197] 22.848270 27.132666 16.035921 21.653186 14.053377 18.901227 20.003557
## [204] 17.710804 21.846118 29.340612 29.768048 24.916565 27.113268 32.706635
## [211] 13.037751 19.952393 21.061998 23.137327 31.729121 17.169715 33.085916
## [218] 26.167766 25.730787 27.436755 20.422723 26.031376 25.319099 20.132472
## [225] 18.873822 17.033086 24.371392 22.866023 27.760461 25.011088 28.325648
## [232] 23.450039 31.351479 23.913499 22.916098 22.831703 21.727520 30.190185
## [239] 29.093626 29.466707 29.116777 23.266323 19.804674 26.399381 30.135126
## [246] 13.691387 26.061734 36.483224 27.701456 22.625983 23.242497 31.616622
## [253] 29.116175 24.755246 20.177139 22.728008 27.491391 17.523578 18.326764
## [260] 16.035947 29.409029 28.480686 22.441846 20.168094 29.744463 13.406936
## [267] 26.521135 19.848240 20.157769 24.104674 26.484234 11.935175 19.754228
## [274] 26.955708 16.472289 33.616459 34.752223 19.693262 30.404273 23.281462
## [281] 17.517113 22.910455 29.105032 27.682235 20.912480 22.309219 27.962531
## [288] 23.758590 24.652303 36.577772 25.534272 23.889227 21.434632 23.183871
## [295] 27.997399 23.131056 19.157312 16.360413 22.613201 31.199142 30.778578
## [302] 17.184911 30.616295 30.854296 18.189519 22.433316 24.218769 29.060366
## [309] 26.943783 28.800496 24.279477 23.066109 18.906900 12.215564 22.607358
## [316] 19.367116 28.278699 19.297931 25.490530 24.981551 25.710395 22.046787
## [323] 22.299137 25.809404 19.106136 26.176935 20.304256 21.290579 26.311361
## [330] 30.493385 26.198492 24.618954 30.635656 24.324071 25.904664 18.510005
## [337] 14.860061 22.749031 17.343776 32.237737 21.901684 16.565161 19.012145
## [344]  5.200234 28.117542 32.244798 19.973812 25.162919 15.685909 16.755559
```

```
plot(z1)
```



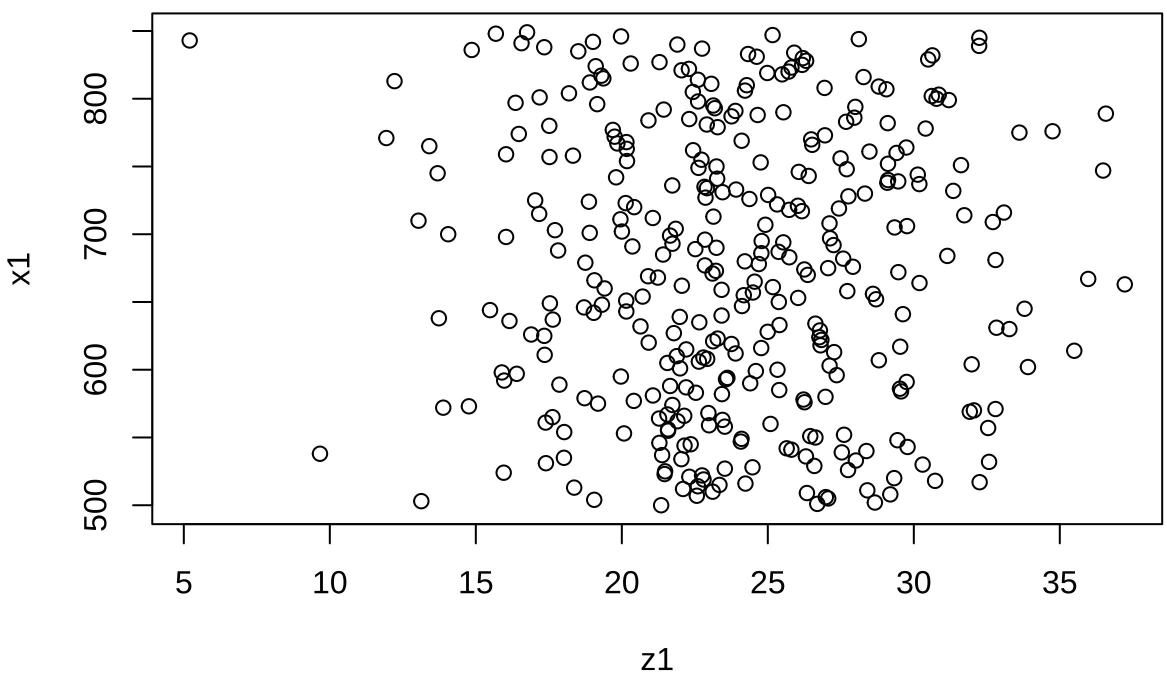
```r
w1<-length(z1)
w1
```

```
## [1] 350
```

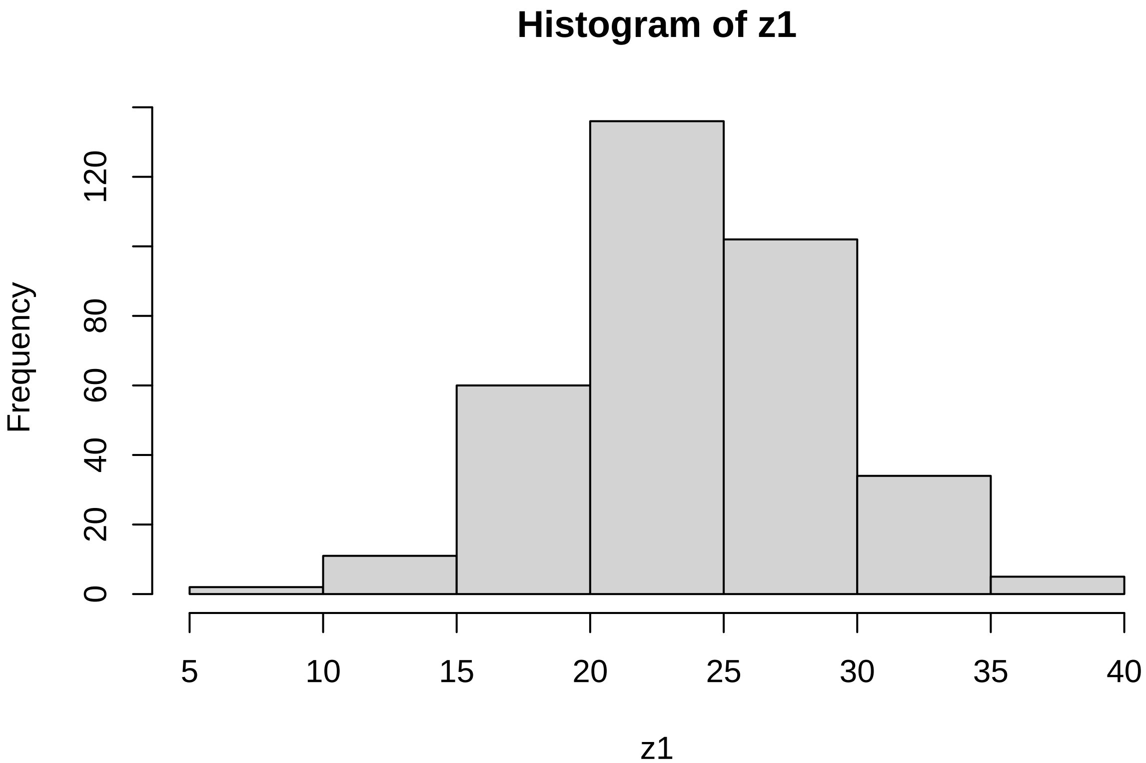
```r
x1<-(500:849)
x1
```

```
##   [1] 500 501 502 503 504 505 506 507 508 509 510 511 512 513 514 515 516 517
##  [19] 518 519 520 521 522 523 524 525 526 527 528 529 530 531 532 533 534 535
##  [37] 536 537 538 539 540 541 542 543 544 545 546 547 548 549 550 551 552 553
##  [55] 554 555 556 557 558 559 560 561 562 563 564 565 566 567 568 569 570 571
##  [73] 572 573 574 575 576 577 578 579 580 581 582 583 584 585 586 587 588 589
##  [91] 590 591 592 593 594 595 596 597 598 599 600 601 602 603 604 605 606 607
## [109] 608 609 610 611 612 613 614 615 616 617 618 619 620 621 622 623 624 625
## [127] 626 627 628 629 630 631 632 633 634 635 636 637 638 639 640 641 642 643
## [145] 644 645 646 647 648 649 650 651 652 653 654 655 656 657 658 659 660 661
## [163] 662 663 664 665 666 667 668 669 670 671 672 673 674 675 676 677 678 679
## [181] 680 681 682 683 684 685 686 687 688 689 690 691 692 693 694 695 696 697
## [199] 698 699 700 701 702 703 704 705 706 707 708 709 710 711 712 713 714 715
## [217] 716 717 718 719 720 721 722 723 724 725 726 727 728 729 730 731 732 733
## [235] 734 735 736 737 738 739 740 741 742 743 744 745 746 747 748 749 750 751
## [253] 752 753 754 755 756 757 758 759 760 761 762 763 764 765 766 767 768 769
## [271] 770 771 772 773 774 775 776 777 778 779 780 781 782 783 784 785 786 787
## [289] 788 789 790 791 792 793 794 795 796 797 798 799 800 801 802 803 804 805
## [307] 806 807 808 809 810 811 812 813 814 815 816 817 818 819 820 821 822 823
## [325] 824 825 826 827 828 829 830 831 832 833 834 835 836 837 838 839 840 841
## [343] 842 843 844 845 846 847 848 849
```

```r
plot(z1,x1)
```



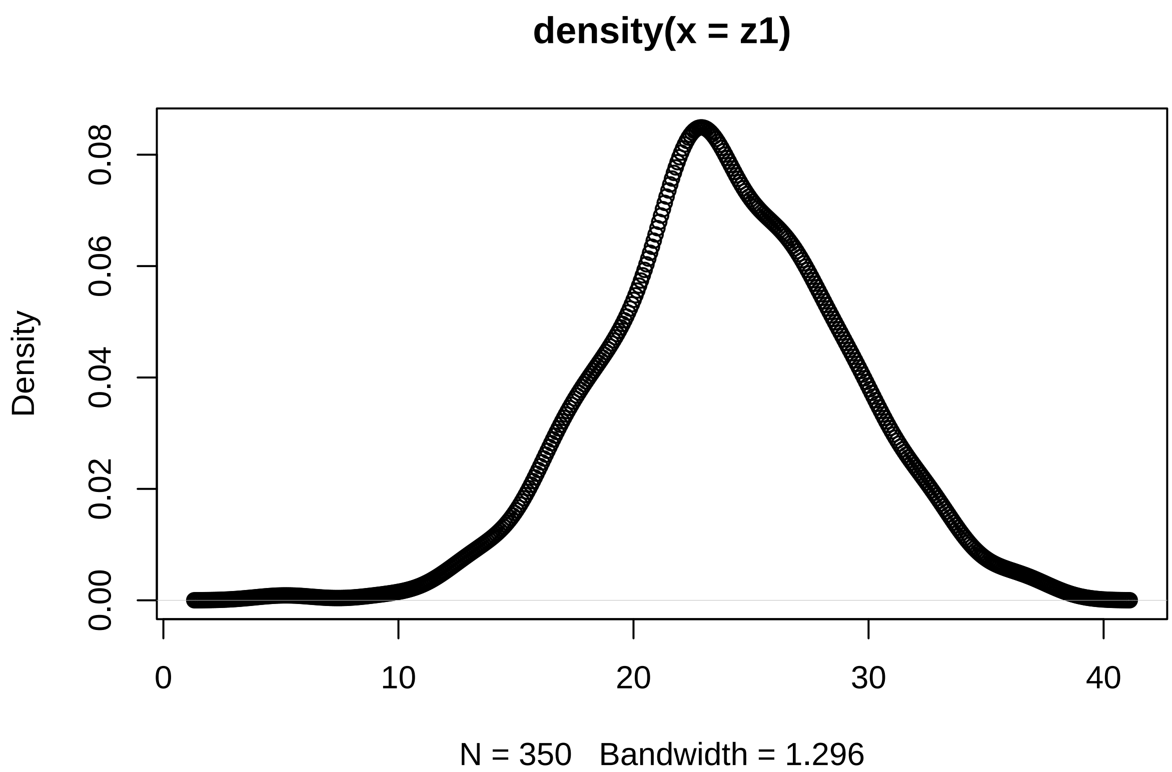
```r
hist(z1)
```




```r
density(z1) #estadísticas de los cuantiles
```

```
##
## Call:
##  density.default(x = z1)
##
## Data: z1 (350 obs.); Bandwidth 'bw' = 1.296
##
##        x                 y
##  Min.   : 1.314   Min.   :9.862e-06
##  1st Qu.:11.263   1st Qu.:8.822e-04
##  Median :21.213   Median :1.164e-02
##  Mean   :21.213   Mean   :2.508e-02
##  3rd Qu.:31.163   3rd Qu.:4.614e-02
##  Max.   :41.112   Max.   :8.491e-02
```

```r
plot(density(z1),type="p")
```


density(x = z1)
Density
0.00
0.02
0.04
0.06
0.08
0
10
20
30
40
N = 350   Bandwidth = 1.296